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THE EFFECTIVENESS OF BRAIN GYM EXERCISES IN IMPROVING COGNITIVE FUNCTION IN OLDER ADULTS : LITERATURE REVIEW

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Abstract

Background: Cognitive decline is common among older adults and may reduce independence and quality of life. *Brain Gym* is a non-pharmacological intervention used to stimulate brain function and maintain cognitive performance, but evidence of its effectiveness remains varied.

Method: This study employed a *systematic review* design using articles published between 2022 and 2025. Literature searches were conducted through PubMed, ScienceDirect, Google Scholar, DOAJ, Garuda, and SINTA using relevant keywords and Boolean operators. Article selection followed the PRISMA 2020 guidelines and the PICOS framework. A total of 82 articles were identified, of which 10 met the inclusion criteria and were included in the final analysis.

Results: Most of the reviewed studies demonstrated that *Brain Gym* exercises significantly improved cognitive function in older adults, particularly in the domains of memory, attention, concentration, orientation, and executive function. Eight of the ten included studies reported significant improvements in cognitive assessment scores, especially using the *Mini Mental State Examination* (MMSE), following regular *Brain Gym* interventions. However, two international review studies indicated that although individual studies generally showed positive findings, the overall level of evidence remains limited because of methodological heterogeneity, small sample sizes, and variations in intervention protocols.

Conclusion: *Brain Gym exercises may be a safe, effective, and low-cost intervention for improving cognitive function in older adults. Further high-quality randomized controlled trials are needed to confirm their effectiveness*.

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INTRODUCTION

Increasing life expectancy is one of the indicators of successful health development occurring in many countries, including Indonesia. The increase in life expectancy has led to a growing older adult population, resulting in an increased prevalence of health problems associated with the aging process. One of the most common health problems experienced by older adults is cognitive decline, which refers to a decrease in the ability to think, remember, concentrate, understand information, make decisions, and perform various daily activities. Progressive cognitive decline may develop into mild cognitive impairment (MCI) and eventually dementia, leading to decreased independence, reduced quality of life, and increased need for long-term healthcare services (World Health Organization, 2021).

Cognitive impairment in older adults has currently become one of the major public health challenges worldwide. The aging process causes physiological changes in the central nervous system, including a reduction in the number of neurons, decreased synaptic plasticity, impaired communication between nerve cells, and reduced blood flow to brain tissue. These changes gradually impair memory, attention, orientation, language, and executive functions. In addition to the aging process, risk factors such as hypertension, diabetes mellitus, cardiovascular disease, depression, physical inactivity, insufficient mental stimulation, and social isolation may accelerate cognitive decline in older adults (Varela et al., 2023).

In Indonesia, the increasing number of older adults has also been accompanied by an increasing prevalence of cognitive impairment. This condition is concerning because cognitive decline affects not only the ability to think and remember but also the ability of older adults to perform daily activities independently. Older adults with cognitive impairment tend to experience difficulties in communication, medication management, and social interaction and have a higher risk of becoming dependent on family members or caregivers. Therefore, early prevention and intervention are essential to maintain cognitive function and improve the quality of life of older adults (Syukriyah, 2024).

Cognitive function refers to the brain's ability to receive, process, store, and utilize information to perform various activities of daily living. These functions include short-term and long-term memory, attention, orientation, language abilities, visuospatial function, calculation skills, and executive functions. Cognitive decline in older adults is closely associated with structural changes in the brain, particularly in areas such as the hippocampus, prefrontal cortex, and cerebral cortex, which play important roles in learning, memory storage, and decision-making. Declining function in these areas causes learning and memory abilities to become progressively slower with advancing age (Setiawan et al., 2024).

Various approaches have been developed to maintain and improve cognitive function in older adults. In addition to pharmacological therapy, non-pharmacological interventions are increasingly being developed because they are relatively safe, easy to implement, affordable, and associated with minimal side effects. Various forms of physical activity, including aerobic exercise, older-adult exercise programs, yoga, tai chi, balance training, cognitive stimulation, educational games, and motor coordination exercises, have been reported to improve brain health by increasing cerebral perfusion, enhancing brain glucose metabolism, and stimulating brain-derived neurotrophic factor (BDNF), which plays an important role in neuroplasticity and the formation of connections between neurons (Lina et al., 2021).

One form of non-pharmacological exercise that has increasingly been implemented is Brain Gym. Brain Gym is a

series of simple movements developed by Paul E. Dennison based on the concept of Educational Kinesiology. This exercise aims to improve coordination between the right and left hemispheres of the brain through cross-lateral movements, eye coordination, and coordination of the head, hands, and lower extremities. These movements are believed to improve communication between the two cerebral hemispheres, enhance sensorimotor coordination, increase concentration, strengthen memory, and optimize learning abilities through the simultaneous activation of various areas of the brain (Yusuf et al., 2010).

Physiologically, Brain Gym is believed to work by increasing blood circulation and oxygen delivery to brain tissue, stimulating cerebral cortex activity, enhancing interhemispheric coordination, and promoting the release of various neurotrophic growth factors involved in the formation of new synapses. Regular physical activity can also improve nervous system flexibility, enhance vascular function, and maintain the integrity of brain tissue, thereby potentially slowing the decline in cognitive function associated with aging. Therefore, Brain Gym has increasingly been developed as a form of exercise that can be easily implemented in community health services and geriatric care settings (Rahmayati et al., 2024).

Various studies conducted in Indonesia have indicated that Brain Gym exercises may provide benefits for improving cognitive function in older adults. A quasi-experimental study involving older adults with dementia demonstrated that Brain Gym exercises performed regularly over several sessions were able to increase Mini-Mental State Examination (MMSE) scores, improve orientation abilities, enhance memory, and improve attention and concentration among participants. Nevertheless, international research has indicated that the effectiveness of Brain Gym requires further investigation through experimental studies with stronger methodologies and larger sample sizes to obtain more robust conclusions regarding its benefits for cognitive function in older adults (Mendrofa et al., 2020).

In recent years, Brain Gym exercises have increasingly been implemented as a non-pharmacological intervention to maintain and improve cognitive function in older adults. This intervention has attracted attention because its movements are simple, easy to learn, require no special equipment, are inexpensive, and can be performed individually or in groups within community settings. Brain Gym is also considered suitable for implementation in primary healthcare services, older-adult health posts, social care institutions, and other healthcare facilities because it can be performed by older adults with varying levels of physical ability. Furthermore, this exercise is relatively safe and has a low risk of injury when performed according to appropriate procedures. Nevertheless, the effectiveness of Brain Gym in improving cognitive function remains under debate because research findings have demonstrated considerable variation (Varela et al., 2023).

Various studies in Indonesia have reported promising results regarding the use of Brain Gym to improve cognitive function in older adults. A study conducted by Ramayanti et al. (2022) demonstrated that Brain Gym administered to older adults with a history of dementia significantly improved cognitive function based on pre- and post-intervention measurements. The study reported that most participants experienced improvements in orientation, attention, and memory after participating in Brain Gym exercises regularly. Statistical analysis demonstrated significant results supporting the effectiveness of the intervention, suggesting that Brain Gym may be recommended as a complementary therapy in gerontological nursing care (Ramayanti et al., 2022).

Another study conducted by Nurli et al. (2024) reported similar findings. Using a pre-experimental one-group pretest-posttest design, the study demonstrated an

improvement in cognitive function after older adults participated in Brain Gym exercises regularly. Movements involving coordination between both sides of the body are believed to enhance communication between the right and left hemispheres of the brain, thereby improving the ability to think, remember, and maintain concentration. These findings further support the assumption that simple physical activities performed consistently can provide positive stimulation for brain function in older adults (Nurli et al., 2024).

The findings of Azizah et al. (2025) demonstrated that Brain Gym was more effective than Puzzle Therapy in improving cognitive function in older adults, particularly in the delayed recall domain. The study employed a quasi-experimental design involving comparisons between two intervention groups, providing evidence that Brain Gym may improve not only overall memory function but also the ability to recall information after a certain period of time. These findings indicate that Brain Gym may serve as an alternative non-pharmacological intervention that can be easily implemented in community healthcare settings (Azizah et al., 2025).

In addition to primary studies, several literature reviews and meta-analyses have been conducted to evaluate the effectiveness of Brain Gym in improving cognitive function. A meta-analysis conducted by Setiawan et al. (2024) demonstrated that, in general, Brain Gym has the potential to produce positive effects on cognitive function in older adults. However, the magnitude of the observed effects varied considerably and was influenced by differences in participant characteristics, intervention duration, exercise frequency, cognitive assessment instruments, and methodological quality across studies. The high degree of heterogeneity among studies resulted in inconsistent findings, indicating the need for further research using stronger study designs (Setiawan et al., 2024).

Different findings were reported in a systematic review and meta-analysis conducted by Varela et al. (2023), which evaluated ten studies examining Brain Gym among older adults. Although most individual studies demonstrated improvements in cognitive function scores following Brain Gym interventions, the meta-analysis showed that increases in Mini-Mental State Examination (MMSE) scores in the Brain Gym groups were not significantly greater than those observed in the control groups. The authors concluded that the available scientific evidence was of low to moderate quality. Therefore, further studies using randomized controlled trials (RCTs), larger sample sizes, and more standardized intervention protocols are required to obtain stronger evidence regarding the effectiveness of Brain Gym in older adults (Varela et al., 2023).

In addition to Brain Gym, various studies have demonstrated that physical activity in general is closely associated with improved cognitive function in older adults. Physical activity can increase cerebral blood flow, stimulate the release of brain-derived neurotrophic factor (BDNF), improve synaptic plasticity, reduce chronic inflammatory processes, and increase gray matter volume in several brain regions involved in memory and executive functions. These mechanisms explain why physical exercise represents an important strategy for maintaining brain health and slowing age-related cognitive decline (Karamacoska et al., 2023).

Although numerous studies have reported the benefits of Brain Gym for cognitive function in older adults, several research gaps remain. First, most studies have employed pre-experimental or quasi-experimental designs with relatively small sample sizes, limiting the generalizability of their findings. Second, considerable variation exists in the duration, frequency, and types of Brain Gym movements used, making comparisons between studies difficult. Third, the instruments used to assess cognitive function also vary, including the Mini-Mental State Examination (MMSE), Montreal

Cognitive Assessment (MoCA), Short Portable Mental Status Questionnaire (SPMSQ), and other assessment tools. These differences contribute to inconsistencies in research findings and indicate the need for a more comprehensive synthesis of scientific evidence (Lina & Kurniawan, 2022).

Based on the foregoing discussion, Brain Gym can be considered a non-pharmacological intervention with the potential to help maintain and improve cognitive function in older adults. However, the available research findings remain inconsistent, highlighting the need for a comprehensive literature review to analyze, compare, and synthesize scientific evidence regarding the effectiveness of Brain Gym exercises in improving cognitive function among older adults. Therefore, this study aims to examine published research findings on the effectiveness of Brain Gym exercises in improving cognitive function in older adults, thereby providing a scientific basis for the development of gerontological nursing interventions and evidence-based healthcare practices.

METHOD

This study employed a *systematic review* method, which is a research approach conducted systematically to identify, select, critically evaluate, and synthesize scientific evidence from previously published studies in order to answer the research question regarding the effectiveness of Brain Gym exercises in improving cognitive function among older adults. This method was selected because it provides a comprehensive overview of the available evidence, identifies consistencies and discrepancies among study findings, and generates stronger conclusions through the synthesis of scientific evidence.

A systematic literature search was conducted using several reputable electronic databases, namely PubMed, ScienceDirect, Google Scholar, DOAJ, Garuda, and SINTA. The search process employed combinations of keywords using Boolean operators (*AND* and *OR*) to identify articles relevant to the research objective. The keywords used included "*Brain Gym*" OR "*Brain Gym Exercise*" OR "*Brain Exercise*" AND "*Cognitive Function*" OR "*Cognitive Impairment*" OR "*Cognition*" AND "*Older Adults*" OR "*Elderly*" OR "*Aging*" OR "*Lansia*". In addition, searches were conducted using Indonesian keywords, including "senam otak," "*Brain Gym*," "fungsi kognitif," "peningkatan fungsi kognitif," and "lansia" to identify relevant national publications.

The selected articles consisted of scientific publications in either Indonesian or English published between 2020 and 2025. This publication period was selected to ensure that the evidence included in the review reflected recent developments in research concerning the effectiveness of Brain Gym exercises on cognitive function in older adults. All retrieved articles were subsequently screened based on their titles, abstracts, and full-text content to determine their relevance to the research objectives.

The article selection process followed the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020* guidelines. The selection process consisted of several stages, including identification of articles from all selected databases, removal of duplicate records, screening based on titles and abstracts, assessment of eligibility through full-text review, and determination of the articles that met the inclusion criteria for analysis. The use of the PRISMA guidelines was intended to improve the transparency and quality of reporting in this systematic review and to ensure that the article selection process was systematic and reproducible by other researchers.

The inclusion criteria were as follows: (1) original research articles, systematic reviews, or meta-analyses; (2) studies examining the effectiveness or effect of Brain Gym exercises on cognitive function in older adults; (3) participants aged ≥60 years or meeting the definition of older adults used in the respective studies; (4) articles available in full-text format; (5) articles published in

indexed national or international journals; and (6) studies using validated cognitive assessment instruments, such as the *Mini-Mental State Examination (MMSE)*, *Montreal Cognitive Assessment (MoCA)*, *Short Portable Mental Status Questionnaire (SPMSQ)*, or other validated cognitive assessment tools.

The exclusion criteria were as follows: (1) articles that did not examine Brain Gym as the primary intervention; (2) editorials, commentaries, conference abstracts, case reports, or proceedings; (3) articles for which the full text was unavailable; (4) studies involving participants who were not older adults; and (5) articles published outside the specified publication period. These criteria were established to ensure that the studies included in the analysis had an appropriate level of quality and relevance to the research objectives.

Article selection was also guided by the *PICOS (Population, Intervention, Comparison, Outcomes, and Study Design)* framework. The *Population* consisted of older adults experiencing cognitive decline as well as older adults with normal cognitive function. The *Intervention* was Brain Gym or brain exercise. The *Comparison* included control groups, alternative interventions, or pre-intervention conditions. The *Outcomes* assessed changes or improvements in cognitive function based on measurements obtained using validated cognitive assessment instruments. The eligible *Study Designs* included randomized controlled trials, quasi-experimental studies, pre-experimental studies, cross-sectional studies, systematic reviews, and meta-analyses.

All articles that met the inclusion criteria underwent data extraction using a data extraction form developed by the researchers. The extracted data included authors' names, publication year, country of study, research design, sample size, participant characteristics, type of Brain Gym intervention, intervention duration, cognitive assessment instruments, research findings, and study conclusions. The extracted data were subsequently analyzed using a *narrative synthesis* by comparing the characteristics, methodologies, and findings of the included studies. This process was conducted to obtain a comprehensive overview of the effectiveness of Brain Gym exercises in improving cognitive function among older adults.

The literature search identified a total of 82 articles from the selected electronic databases. Following the removal of duplicate records, screening based on titles and abstracts, full-text eligibility assessment, and application of the inclusion and exclusion criteria, 10 articles met the eligibility criteria and were included in this systematic review. The final 10 articles consisted of five national journals and five international journals that examined the effectiveness of Brain Gym exercises in improving cognitive function among older adults and served as the basis for the synthesis of findings in this review.

RESULT

Article Selection Flow Diagram Based on PRISMA 2020

The literature search was conducted using six electronic databases: PubMed, ScienceDirect, Google Scholar, DOAJ, Garuda, and SINTA. The search process using the predetermined keywords identified a total of 82 articles. After removing 18 duplicate articles, 64 articles remained and were subsequently screened based on their titles and abstracts. A total of 41 articles were excluded because they were not relevant to the research topic, leaving 23 articles eligible for full-text assessment. Following the full-text evaluation, 13 articles were excluded because they did not meet the inclusion criteria, including studies that did not use Brain Gym as the primary intervention, did not involve older adult participants, or did not specifically report cognitive function outcomes. Consequently, 10 articles met all the inclusion criteria and were included in the analysis of this systematic review in accordance with the PRISMA 2020 guidelines.

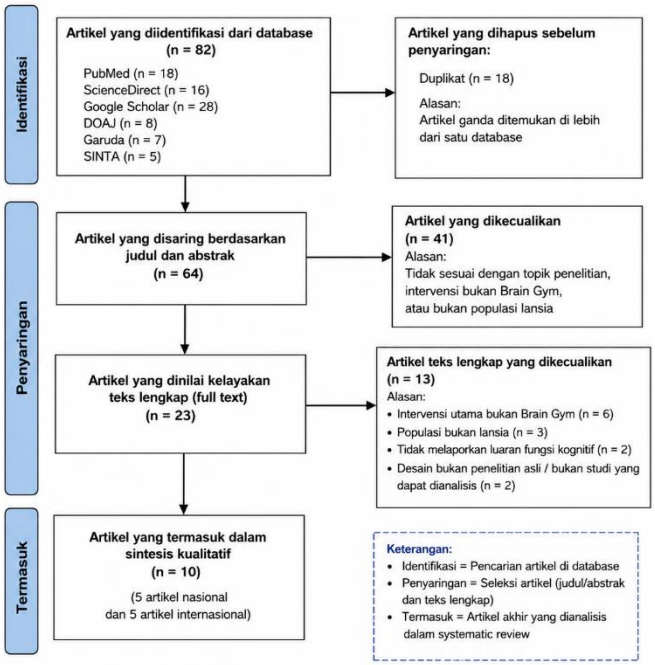


Figure 1. Article Selection Flow Diagram Based on PRISMA 2020

A. Characteristics of the Included Studies

a. Distribution of Publication Years

Table 1. Distribution of Articles by Publication Year

Publication Year	Number of Articles	Percentage
2022	2	20%
2023	2	20%
2024	5	50%
2025	1	10%
Total	10	100%

Based on Table 1, the largest proportion of the analyzed articles was published in 2024, accounting for 5 articles (50%). Articles published in 2022 and 2023 each accounted for 2 articles (20%), while 1 article (10%) was published in 2025. These findings indicate an increasing interest in research examining the effectiveness of Brain Gym in improving cognitive function among older adults in recent years.

b. Distribution by Journal Origin

Table 2. Distribution of Articles by Journal Origin

Journal Origin	Number of Articles	Percentage
National	5	50%
International	5	50%
Total	10	100%

Based on Table 2, the articles included in this systematic review consisted of 5 national journals (50%) and 5 international journals (50%). Thus, the synthesis provides a balanced representation of scientific evidence from both national and international research.

c. Distribution by Database

Table 3. Distribution of Articles by Database

Database	Number of Articles	Percentage
Google Scholar	4	40%
PubMed	2	20%
ScienceDirect	2	20%
SINTA	1	10%
DOAJ	1	10%
Total	10	100%

Based on Table 3, the analyzed articles were obtained from five scientific databases. Google Scholar contributed the largest number of articles, with 4 articles (40%), followed by PubMed and ScienceDirect with 2 articles each (20%). SINTA and DOAJ each contributed 1 article (10%).

d. Distribution by Study Design

Table 4. Distribution of Articles by Study Design

Study Design	Number of Articles	Percentage
Pre-experimental	3	30%
Quasi-experimental	3	30%
Cross-sectional	1	10%
Systematic review	2	20%
Meta-analysis	1	10%
Total	10	100%

Based on Table 4, the most frequently used study designs were pre-experimental and quasi-experimental designs, each accounting for 3 articles (30%). In addition, the review included 2 systematic reviews (20%), 1 meta-analysis (10%), and 1 cross-sectional study (10%).

e. Distribution by Cognitive Function Assessment Instruments

Table 5. Distribution of Cognitive Function Assessment Instruments

Assessment Instrument	Number of Articles	Percentage
MMSE	6	60%
MoCA	2	20%
SPMSQ	1	10%
Literature Review/Meta-analysis	1	10%
Total	10	100%

Based on Table 5, the most frequently used assessment instrument was the Mini-Mental State Examination (MMSE), which was used in 6 articles (60%). Other instruments included the Montreal Cognitive Assessment (MoCA), used in 2 articles (20%), and the Short Portable Mental Status Questionnaire (SPMSQ), used in 1 article (10%). One article was a meta-analysis and therefore did not directly use a cognitive assessment instrument.

f. Distribution of Research Findings

Table 6. Distribution of Research Findings

Research Finding	Number of Articles	Percentage
Brain Gym significantly improved cognitive function	8	80%
Brain Gym showed improvement, but the evidence was inconsistent	2	20%
Total	10	100%

Based on Table 6, 8 articles (80%) reported that Brain Gym exercises significantly improved cognitive function among older adults. Meanwhile, 2 articles (20%), consisting of an international systematic review and meta-analysis, reported that although most individual studies demonstrated positive findings, the overall scientific evidence remained inconsistent. These findings indicate the need for further research using stronger methodological approaches.

g. Distribution of Cognitive Domains Showing Improvement

Table 7. Cognitive Domains Reported to Have Improved

Cognitive Domain	Number of Articles	Percentage*
Memory	8	80%
Attention	7	70%
Concentration	7	70%
Orientation	5	50%
Executive function	4	40%

Based on Table 7, memory was the cognitive domain most frequently reported to improve following Brain Gym exercises, as reported in 8 articles. Improvements were also frequently observed in attention, concentration, orientation, and executive function. These findings suggest that Brain Gym may provide benefits across multiple domains of cognitive functioning in older adults.

*Percentages in Table 7 are calculated based on the total of 10 included articles. Because one study could report improvements in more than one cognitive domain, the percentages are not mutually exclusive.

h. Synthesis of Research Findings

Based on the synthesis of the 10 analyzed articles, most studies demonstrated that Brain Gym exercises had a positive effect on cognitive function among older adults. Improvements were generally reflected in increased Mini-Mental State Examination (MMSE) or Montreal Cognitive Assessment (MoCA) scores following regular Brain Gym exercises. The cognitive domains most frequently reported to improve included memory, attention, concentration, orientation, and executive function. However, two international studies employing systematic review and meta-analysis designs reported that although most individual studies demonstrated positive outcomes, the overall quality of the available evidence remained low to moderate due to methodological heterogeneity, variations in intervention duration, relatively small sample sizes, and differences in cognitive assessment instruments. Therefore, further randomized controlled trials (RCTs) with stronger methodological designs are needed to establish more robust evidence regarding the effectiveness of Brain Gym in improving cognitive function among older adults.

DISCUSSION

The results of the systematic review of 10 articles demonstrated that most studies reported improvements in cognitive function among older adults following Brain Gym exercises. Eight of the 10 analyzed articles showed improvements in cognitive function scores following Brain Gym intervention, whereas two articles, consisting of a systematic review and a meta-analysis, indicated that although most individual studies reported positive findings, the overall scientific evidence requires further investigation using stronger methodological approaches. These findings suggest that Brain Gym has potential as a non-pharmacological intervention for maintaining and improving cognitive function in older adults, particularly when implemented regularly and systematically (Varela et al., 2023).

Improvements in cognitive function following Brain Gym exercises were observed across several domains, including memory, attention, concentration, orientation, and executive function. Most of the studies used the Mini-Mental State Examination (MMSE) to assess changes in

cognitive function and reported increased scores following the intervention. These findings indicate that simple physical activities involving body coordination may provide stimulation to brain activity and consequently improve thinking and memory abilities in older adults. This result is consistent with research conducted in Indonesia, which reported that regular Brain Gym exercises significantly increased MMSE scores after several weeks of intervention (Nurli et al., 2024).

Physiologically, Brain Gym exercises are proposed to stimulate coordination between the right and left hemispheres of the brain. Cross-lateral movements, balance exercises, eye coordination, and bilateral motor activities may enhance communication between the two cerebral hemispheres through the corpus callosum. These activities may also increase blood flow to brain tissue, thereby optimizing the delivery of oxygen and glucose to the brain. Such conditions may support neuroplasticity, which refers to the brain's ability to form and strengthen connections between nerve cells that are important for learning, memory, and cognitive function. Therefore, Brain Gym may be considered a form of sensorimotor stimulation that can potentially help maintain brain function during the aging process (Dennison & Dennison, 2010).

In addition to improving cerebral perfusion, light physical activity such as Brain Gym may stimulate the release of neurotrophic factors, including brain-derived neurotrophic factor (BDNF). Increased BDNF levels play an important role in neuronal survival, synaptic strengthening, learning, and memory. In older adults, reduced BDNF levels have been associated with age-related cognitive decline. Therefore, regular Brain Gym exercises may potentially contribute to maintaining cognitive function through these biological mechanisms (Erickson et al., 2019).

The national studies included in the review generally demonstrated consistent positive effects of Brain Gym on cognitive function among older adults. Research conducted by Ramayanti et al. reported improvements in cognitive function among older adults with a history of dementia following regular Brain Gym exercises. Similarly, research by Karim and Puspita reported significant improvements in cognitive function scores following the intervention, while a study by Satria et al. demonstrated that the addition of Brain Gym exercises increased MMSE scores among older adults. The consistency of these findings supports the potential application of Brain Gym as a non-pharmacological intervention in community-based healthcare and geriatric services (Ramayanti et al., 2022).

Nevertheless, international systematic reviews and meta-analyses have indicated that the quality of scientific evidence concerning the effectiveness of Brain Gym remains variable. This variation may be attributed to differences in study designs, relatively small sample sizes, inconsistent intervention durations, and the use of different cognitive assessment instruments. Such methodological differences make direct comparisons between studies difficult and may contribute to heterogeneity in the observed effects of Brain Gym. Therefore, several researchers have recommended the use of randomized controlled trials (RCTs) with larger sample sizes to establish stronger evidence regarding its effectiveness (Setiawan et al., 2024).

Differences between national and international findings may also be influenced by participant characteristics. Most national studies involved older adults with mild to moderate cognitive impairment living in the community or social care facilities, whereas some international studies included more heterogeneous populations, including cognitively healthy older adults and individuals with dementia at different levels of severity. In addition, the frequency of Brain Gym exercises varied among studies, ranging from two to five sessions per week, with different intervention durations. These differences in participant characteristics and intervention protocols may influence the

magnitude of the intervention's effect on cognitive function (Lina & Kurniawan, 2022).

From a nursing practice perspective, Brain Gym has several advantages compared with other non-pharmacological interventions. The exercises are relatively simple, require no specialized equipment, are easy for older adults to learn, can be adapted to different levels of physical ability, and can be implemented in community health centers, older-adult health posts, social care facilities, and family settings. In addition, the relatively low implementation cost makes Brain Gym a potentially useful promotive and preventive intervention for maintaining cognitive health among older adults. Therefore, Brain Gym may be considered as part of community-based geriatric healthcare programs (Mahfud & Jelita, 2023).

Despite the generally positive findings, this systematic review has several limitations. The number of articles meeting the inclusion criteria was relatively small, and substantial variations were observed in study designs, sample sizes, intervention durations, and cognitive assessment instruments. Furthermore, most studies employed pre-experimental or quasi-experimental designs, resulting in a lower level of evidence compared with randomized controlled trials. These limitations should be considered when interpreting the findings, and further research using stronger methodological designs and more standardized intervention protocols is warranted (Page et al., 2021).

Overall, the findings of this systematic review indicate that Brain Gym exercises have potential as a non-pharmacological intervention for improving cognitive function in older adults. Most studies reported improvements in memory, attention, concentration, orientation, and executive function following regular Brain Gym exercises. Nevertheless, the quality of the available scientific evidence requires further strengthening through experimental studies with more rigorous research designs. Therefore, Brain Gym may be considered a promotive and preventive strategy for maintaining cognitive function and improving the quality of life of older adults (Karim & Puspita, 2024).

CONCLUSION

Based on the systematic review of 10 articles that met the inclusion criteria, Brain Gym exercises have the potential to improve cognitive function among older adults. Most of the reviewed studies demonstrated improvements in memory, attention, concentration, orientation, and executive function following regular Brain Gym exercises. However, several international systematic reviews and meta-analyses indicated that the quality of the available scientific evidence remains variable due to differences in study designs, sample sizes, intervention durations, and cognitive assessment instruments. Therefore, Brain Gym may be considered a simple, safe, and accessible non-pharmacological intervention to help maintain and improve cognitive function among older adults.

RECOMMENDATIONS

Brain Gym exercises are recommended for integration into older-adult healthcare programs at community health centers, older-adult health posts, social care facilities, and community settings as a promotive and preventive strategy for maintaining cognitive function. Future research should employ randomized controlled trial (RCT) designs with larger sample sizes, longer intervention periods, and standardized cognitive assessment instruments to provide stronger scientific evidence regarding the effectiveness of Brain Gym exercises in improving cognitive function among older adults.

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